J. Edgar Hernandez Cancino Estrada

Robotics & ML Engineer / Data Scientist

Portfolio GitHub LinkedIn

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Summary

As a passionate robotic engineering student with a deep interest about machine learning and artificial intelligence, I bring a fusion of computational expertise, hands-on projects and research, with a proven ability for rapid learning and creative problem-solving. I am eager to further develop my skills and contribute to the forefront of robotics and AI advancements.

Education

- **Instituto Tecnologico y de Estudios Superiores de Monterrey (ITESM), Mexico** *Aug 2019 - June 2024*

  B.S. in Robotics and Digital Systems Engineering — Minor-like in Physics Engineering Percentage: 96/100

Research experience

- **Rice University** *Jan 2024 - Feb 2024*

  Winter research internship at Dr. Lydia Kavraki's lab for Computational Robotics and AI. Houston, USA

  – Building of robotics systems and development of scalable and maintainable code.
  – Integration of learned models for perception and grasp pose prediction with motion planning frameworks.

- **ITESM** *Aug 2023 - Dec 2023*

  Semester-based research internship Monterrey, Mexico

  – Individual research stay on application of Deep Reinforcement Learning for trajectory planning of a reaching task for a 6-DOF robotic arm.

Professional experience

- **Lava Tap** *Oct 2022 - Present*

  Data Scientist Monterrey, Mexico

  – Implementation of a machine learning models to estimate the duration of semi-truck loading operations for a tissue paper distribution plant.
  – Development of innovative payment and charging schemes for the company's personnel and clients, utilizing historical data, with an estimated reduction in operational costs of up to 30%.
  – Creation of data visualization dashboards in Tableau for business presentations.

- **Steelcase Inc.** *Mar 2022 - Oct 2023*

  Business Automation Intern Monterrey, Mexico

  – Developed and implemented process automation solutions with technologies such as RPA, Python, and Visual Basic to automate more than 5000 annual hours for internal processes in areas such as finance, operations, and sustainability.

Projects

- **Deep Reinforcement Learning for trajectory planning** *Aug 2023 - Dec 2023*

  Individual research Stay at Monterrey Institute of Technology and Higher Education **Github** | **Research Paper**

  – Utilized Deep Deterministic Policy Gradient (DDPG) and Hindsight Experience Replay (HER) to make a xArm6 robotic arm learn to reach a target position from a starting pose.
  – Implementation of the model in a real UFactory xArm6 robot and in RViz simulation using ROS.

- **Multilayer Perceptron and Support Vector Regression for truck loading time estimation** *Nov 2022 - Sept 2023*

  Business Partner: Lava Tap **Research Paper**

  – Implemented MLP, SVR, and Ridge Regression to estimate the duration of semi-trailer truck loading maneuvers for a tissue paper warehouse in Coahuila, Mexico for a real commercial application.

- **Autonomous vehicle navigation challenge** *Apr 2023 - June 2023*

  Challenge by Manchester Robotics and school project **Video report**

  – Participated in a team developing an autonomous mobile robot using ROS and applied Computer Vision in an Nvidia Jetson Nano device.
  – Implemented linear control strategies for controlling the robot's navigation, and Deep Neural Networks (YOLOv5) for visual identification of transit signs.

- **Visual servoing for xArm6 robot** *June 2023 - June 2023*

  School project **Github**

  – Creation a visual servoing program that enables a robotic arm to track and follow an object identified through a camera and QR code by estimating the target's coordinates relative to the end effector of the robotic arm.
  – Utilized ROS and OpenCV in python, as well as MoveIt for motion planning.

- **Product demand forecast using SARIMA and LSTM** *Feb 2023 - Mar 2023*

Freelance project **View in Portfolio**

  - Implemented a Long Short Term Memory (LSTM) Recurrent Neural Network and the statistcal model SARIMA to estimate product demand on lense frames up to 24 months in advance.

- **Intelligent air pressure control system** *Oct 2022 - Dec 2022*

John Deere training partner school project **View in Portfolio**

  - Built and developed a prototype for an intelligent air pressure control system for tires in agricultural equipment using linear control strategies.
  - Used the ARM cortex processor in a dual core STM32 board. Utilized CAN, and serial communication protocols for communication with sensors and a NodeMCU device for internet connectivity.

- **Music Recognition algorithm in Matlab** *Oct 2020 - Nov 2020*

School project **Github**

  - Developed a music recognition algorithm in MATLAB able to identify a playing song with 10 second samples by applying the Fast Fourier Transform to perform a time-frequency analysis of song files and microphone input, containing up to a dozen million data points.

Technical Skills

- **Programming Languages**: Python, C/C++, MATLAB
- **Technologies**: Keras, Tensorflow, PyTorch, Scikit-learn, Pandas, Tableau, Gym, ROS, OpenCV, SQL, Arduino, MQTT
- **Operating Systems**: Linux, MacOS, Windows

Certifications

- **Machine Learning Specialization**, Stanford & DeepLearning.AI offered through Coursera *2024*
- **Diploma in Data Science**, Technological Institute of Petroleum and Energy, Mexico *2021*

Courses Taken

- **CS50's Introduction to Artificial Intelligence**, HarvardX *2024*
- **Master the Fourier Transform and its Applications**, Udemy *2021*
- **CS50's Introduction to Computer Science**, HarvardX *2020*
- **Building Database Applications in PHP**, University of Michigan through Coursera *2020*
- **Inclusive Education**, Queensland University of Technology *2020*
- **Astrobiology and the Search of Extraterrestial Life**, The University of Edinburgh *2020*

Achievements

- **Scholarship for academic excellence**, Recipient of the highest academic scholarship, at ITESM. *2019*